

Assignment 1: Chapters 1-5

Due 5pm, 13 February 2026

Answer each of the following questions along with a brief explanation (1- 3 sentences) of why it is correct.

1. A person living in a Hong Kong high-rise apartment has a large window with an unobstructed view allowing them to see all locations of the sky that are to the North of East and West (All altitudes from 0° to 90° ; all azimuths from 270° to 360° and from 0° to 90°). During which seasons will they be able to see the sunrise? Around what time of year will they be able to glimpse the sun at noon?
2. Chloe works in Tokyo. Every day, around noon, she walks west from her office to her favorite café for lunch. She enjoys the shade of the buildings that line either side of the street. She notices that during some part of the year, the shadows of the buildings are long enough as to provide shade on both sides of the street, while in other times of the year one side of the street is in the sunlight.
 - a. Which buildings are providing the shade, the ones on the north or south side of the street?
 - b. During which time of year are the shadows longest?
3. Over the course of many nights, a northern observer locates the same star and watches its daily motion. She observes that no matter what time of night or what time of year she always can spot this star in the sky and she never witnesses it rising above or setting below the horizon. In which direction (North, South, East, West) does she face to watch this star?
4. Consider observations of the sun's daily and annual motions. Compare and contrast observations made in Hong Kong and Beijing.
 - a. Compare the noon altitudes of the sun on the Autumnal Equinox. Repeat this comparison for the Winter Solstice.
 - b. Compare the azimuths of sunrise on the Winter Solstice. Repeat this comparison for the Vernal Equinox.
 - c. Compare the lengths of daylight on the Vernal Equinox. Repeat this comparison for the Summer Solstice.
5. In Lecture 3, we learned of several observations that led to ancient Greek astronomers concluding that the Earth is round. If an ancient astronomer only observed the sky from a single location, what evidence, if any, could lead them to adopt a round Earth hypothesis?